

Photosynthesis

The Solar-Powered Kitchen

SCIENCE · CLASS 8 · 45-MINUTE LESSON · CAMBRIDGE / NATIONAL

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Learning Objectives

By the end of this lesson, students will be able to:

- Define photosynthesis and explain its importance for life on Earth.
- Identify the raw materials (carbon dioxide and water) and products (glucose and oxygen) of photosynthesis.
- Write and interpret the word equation for photosynthesis.
- Explain the role of chlorophyll and sunlight in the process.
- Describe the role of leaves as the main site of photosynthesis.

Resources Required

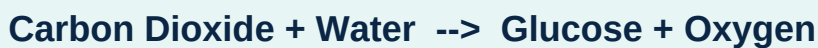
- Whiteboard and markers
- Printed diagram of a leaf cross-section (*included below*)
- Green leaf and yellow leaf for comparison, if available
- Worksheet for the student activity (*included on the last page*)

Lesson Structure

Phase	Time	Activity	Purpose
Starter / Hook	5 min	Ask students: 'If you were locked in a sealed room with a plant and sunlight, would you survive? Why?' Take 2–3 responses. Reveal that plants produce the oxygen we breathe.	Activate curiosity and prior knowledge
Learning Objective	2 min	Write on board: 'By the end of today, you will be able to explain what photosynthesis is, what goes in, and what comes out.'	Set clear expectations
Main Teaching	18 min	1. Draw a leaf on the board — ask what it needs to make food. 2. Introduce chlorophyll; show a green vs. yellow leaf comparison. 3. Write the word equation: carbon dioxide + water --> glucose + oxygen (using sunlight + chlorophyll). 4. Analogy: a plant is a solar-powered kitchen — CO₂ and water are the ingredients, glucose is the food, oxygen is the leftover.	Build conceptual understanding step by step
Student Activity	12 min	Structured worksheet: (a) label the word equation, (b) match raw materials and products, (c) answer 'Why do plants need sunlight?' Pair-share before class discussion.	Reinforce understanding and peer learning

Phase	Time	Activity	Purpose
Wrap Up	8 min	Ask 3 students to each state one thing they learned. Summarize on the board. Exit question: 'Name one raw material and one product of photosynthesis.'	Check understanding and consolidate learning

Key Concept — Word Equation



in the presence of sunlight and chlorophyll

Leaf Cross-Section

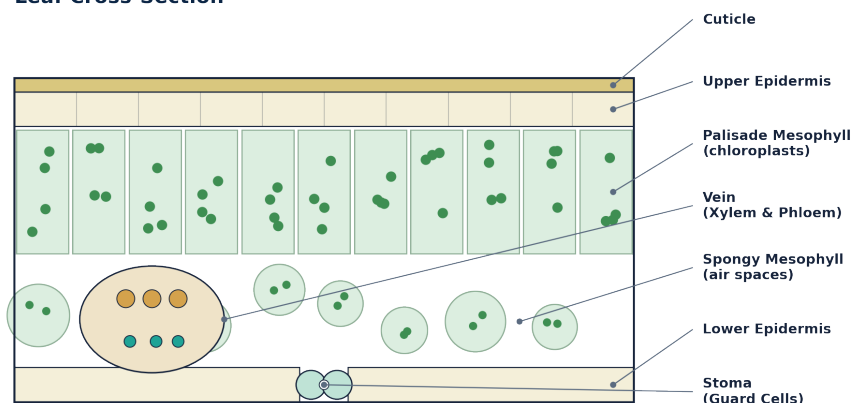


Figure 1. Leaf cross-section — the site of photosynthesis. Palisade mesophyll cells pack the most chloroplasts, which is why they carry out most of the light-capture.

Differentiation

Struggling Students	Average Students	Advanced Students
Provide a partially completed word equation and a simplified diagram to fill in.	Complete the standard worksheet independently and explain the equation to a partner.	Write a short paragraph explaining what would happen to plants if sunlight or CO ₂ were removed.

Assessment

- Observation of student responses during questioning.
- Worksheet completion checked during the activity phase.
- Exit question answered verbally before leaving class.

Teacher Notes

- Use the green vs. yellow leaf comparison to make chlorophyll visually clear.
- The solar-powered kitchen analogy works very well for Class 8 — use it consistently.
- If time runs short, skip the advanced differentiation task.
- Ensure all students copy the word equation correctly into their notebooks.

Student Worksheet

Photosynthesis — Class 8

Name: _____ Class: 8 _____ Date: _____

Part A — Label the Word Equation

Fill in the blanks using the word bank below.

_____ + _____ --> _____ + _____
(in the presence of _____ and _____)

Word bank: Carbon Dioxide · Water · Glucose · Oxygen · Sunlight · Chlorophyll

Part B — Match Raw Materials and Products

Write the correct letter from Column B next to each item in Column A.

Column A	Column B
1. Carbon Dioxide ____	A. The sugar the plant makes and uses as food
2. Water ____	B. Absorbed by the roots from the soil
3. Glucose ____	C. A gas taken in from the air through the stomata
4. Oxygen ____	D. A gas released as a by-product, which we breathe

Part C — Short Answer

Why do plants need sunlight? Write 2–3 sentences.
